

DAVID ALEXANDER HILL

AI Automation & Development

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PROFILE

AI-focused developer with hands-on experience building multi-agent AI systems, LLM-based applications and retrieval-backed memory architectures. Designs system architecture and agent behaviour, then delivers through AI-assisted development — owning the design decisions, code review, debugging and production validation. Self-hosts a personal multi-agent AI stack in daily production use, and led the deployment of a production multi-agent system for a client workforce, including the implementation and integration of MemPalace, an open-source ChromaDB-based memory system exposing 29 tool functions via its MCP server for persistent context across agents. Coding with AI tools since 2022, from early manual prompt-loop workflows through to modern agentic development. Seeking a role where AI development, LLM engineering and automation intersect.

EXPERIENCE

Neozeit Dec 2023 – Present

IT Engineer — Development & Automation

Led the design and deployment of a multi-agent AI automation system (OpenClaw) for Kernel & Flint, including implementing and integrating MemPalace — an open-source ChromaDB-based memory system exposing 29 tool functions via its MCP server — for persistent context across agents.

Designed a tiered model-delegation strategy for cost control across the agent fleet: reasoning models for planning and judgment, lightweight models for mechanical execution, and direct scripts for bulk data work.

Built and deployed production web applications for clients, including a doctor location mapping tool (Node.js) and automated synthetic monitoring for uptime and performance tracking (Python/Playwright).

Developed and maintain a financial market dashboard for Adviceworx with automated Telegram delivery of daily updates and Claude-powered market summarisation, built for Neozeit client work using a custom Telegram bot.

Built a Microsoft 365 recon and visibility dashboard using Extractor Suite and Analyzer Suite outputs, providing deep insight into SharePoint environments and tenant-wide security statistics.

Provide IT support and infrastructure administration across Neozeit's SME client portfolio: second-line Microsoft 365 support (Exchange Online, SharePoint, Teams, Entra ID provisioning and licensing), vulnerability remediation scripting (Python, PowerShell) for ongoing CVE patching, and on-premises Active Directory domain deployment (OU structure, GPO, user management).

Sole operator of Neozeit's Proxmox VE infrastructure (3-node cluster, 12-drive RAID6 backup server, LTO-4 tape rotation); led the Windows Server 2025 migration for NHC (medical client); set up managed switches, VLANs and IP camera/NVR systems including 10GbE SFP+ infrastructure.

Promoted from Junior to IT Engineer after 4 months, recognised for operating consistently above entry level across all project areas.

Drago Technologies Oct 2022 – Feb 2023

IT Technician

Delivered first-line IT support for business clients, resolving hardware, software and connectivity issues.

Assisted with desktop and laptop configuration, OS deployment and peripheral setup across client environments.

TGVision Oct 2021 – Aug 2022

TV Technician

Diagnosed and repaired flat-panel televisions across a range of brands and screen technologies.

Performed backlight array replacement, CoF (Chip on Film) bonding, screen replacement, capacitor replacement and mainboard repair.

Managed repair workflow from intake through diagnosis, parts sourcing, repair execution and quality assurance.

AVeS Cyber Security Jun 2019 – Aug 2019

Intern — Technical Administrator

Provided IT hardware and software troubleshooting support internally and for clients.

Assisted account managers by contacting clients and communicating product and package benefits.

Gained foundational exposure to cybersecurity tooling and the vendor ecosystem, directly informing later Kaspersky and Barracuda certifications.

Earlier Roles 2016 – 2018

Au Pair · Paintball Marshal · Retail Cashier (Part-Time)

Private Family (2018) — daily transport and homework support for a school-age child. Battlezone Paintball (2017–18) — safety briefings, game flow and equipment maintenance. Riverside Liquors (2016–17) — PoS operation, sales and stock queries.

KEY PROJECTS

Personal Multi-Agent AI Stack

Architected and self-host a production multi-agent system on OpenClaw for daily personal and family use: three isolated agents (a primary personal assistant, a sports-science coaching agent used live by a professional cycling coach, and a family-support agent) plus a local model runner (Gemma4 via Ollama) for free on-device inference. Each agent runs its own isolated MemPalace — an open-source ChromaDB-based memory system exposing 29 tool functions via its MCP server — for persistent, agent-scoped memory with zero cross-contamination. Self-hosted end-to-end with zero external dependencies for core execution.

Flint & Kernel Multi-Tenant Agent Deployment

Designed and deployed a production multi-agent system on OpenClaw for a client workforce spanning two tenant companies — an orchestrator, a live production executive-assistant agent, and two department agents (marketing and social media) designed and scoped for rollout. Implemented and integrated MemPalace, an open-source ChromaDB-based memory system exposing 29 tool functions via

its MCP server, with physically separate palace instances and wing-scoped isolation per tenant. Tenant and department boundaries are enforced by filesystem and tool-schema configuration, not just documented policy, backed by a gated knowledge-graph write protocol and automated diary, compression and promotion pipelines. Migrated the agent fleet off the Anthropic API to Kimi K3 mid-2026, cutting inference costs while enhancing reliability, keeping the architecture provider-agnostic by design. Ongoing development and maintenance.

Maropa — AI Document-Intelligence Agent (Upcoming)

Confirmed upcoming engagement, not yet started: an agent that reads Excel spreadsheets, invoices and PDFs and pre-populates Xero accounting entries, reducing manual data capture in bookkeeping workflows.

Charlie Dashboard — AI Cyber Intelligence & GTM Platform (Kernel Afrika)

Full-stack platform combining RSS threat-intel aggregation, autonomous competitor research and persona-tailored content generation. Custom ReAct-style agent loop built directly against the Anthropic Messages API — no agent framework — with end-to-end SSE streaming of agent reasoning and tool calls to a live UI. Deliberately tiered: deterministic keyword heuristics on the high-volume path, model calls reserved for curated work. React, TypeScript, Fastify, Prisma, PostgreSQL. In active development.

Doctor Location Mapping Tool (Node.js)

Production mapping platform for a medical client, enabling directors and marketing teams to visualise which doctors work with the organisation and where they are located. Node.js, Express, Sequelize, MariaDB, Leaflet, with React web, React Native Android and a PWA build. Ships a read-only MCP server exposing codebase query tools.

Advice Works Financial Market Dashboard

Financial market dashboard for Adviceworx, with automated Telegram delivery of daily updates and Claude-powered market summarisation, built for Neozeit client work using a custom Telegram bot.

SELF-DIRECTED DEVELOPMENT (2022 – PRESENT)

Began coding with AI assistance in 2022, using early-generation LLMs in a manual prompt-and-paste workflow before agentic coding tools existed. Now works fluently across modern AI development tooling including Claude Code and Cursor.

Self-studied Python, JavaScript, HTML and CSS through hands-on project work.

Built service health monitoring scripts with automated alerting for personal web services.

Developed data-fetching and categorisation scripts to organise and process structured datasets.

Active GitHub repository spanning personal and professional projects across the full development lifecycle.

SELF-DIRECTED DEVELOPMENT

TECHNICAL SKILLS

Multi-Agent AI Systems (OpenClaw) · Agentic System Architecture · MCP (Model Context Protocol)

LLM Integration & Prompt Engineering · RAG & Vector Databases · ChromaDB / Vector Memory Systems

AI Application Development · Anthropic API / Claude · Python, JavaScript & PowerShell

Node.js & Playwright Automation · HTML, CSS & API Integration · Microsoft 365 & Entra ID

Active Directory (Users, GPO, OUs) · Proxmox VE & Cluster Management · Windows Server & Domain Admin

Managed Switches, VLANs & Networking · Firewall & Network Security · Desktop, Laptop & Server Hardware

EDUCATION

Matric

Cedarwood College, 2016

CERTIFICATIONS

Kaspersky Sales Specialist

S02.11.1 · S41.10 · S40.70 · S20.10 · S39.02 · S38.30

Barracuda Certified Sales Partner

Email Security · Web Security · Vulnerability Remediation · Message Archiver · Load Balancer ADC · DDoS Prevention

LANGUAGES

English — Native

INTERESTS

AI & Automation Development

Computer Hardware & Custom Builds

Skateboarding

Music

Gaming & High-Performance Tech

Content Creation & Video Editing